OSS Observatory
METHODS FOR A MORE STABLE WORLD

# CyberBrief Analytical Variance Framework (AVF)

*Testing the durability of analytical constructs across different execution engines*

SAME METHOD.
MULTIPLE PERSPECTIVES.
DEEPER TRUTH.

## Core Proposition

***"A durable analytical construct should remain meaningfully stable under changes in the execution engine, even when evidence discovery, selection, language, and local interpretation vary."***

The CyberBrief Analytical Variance Framework (AVF) provides a structured way to analyze where and why different execution engines (e.g., Claude, GPT, or others) agree or diverge when applying the same CyberBrief method to the same research problem.

| | 1. Evidence | 2. Selection | 3. Classification | 4. Interpretation | 5. Longitudinal Meaning |
|---|---|---|---|---|---|
| **Analytical Pipeline** *What CyberBrief does at each stage* | Discovers and establishes the factual substrate. | Determines what is worthy of inclusion. | Assigns Signal Type, Half-Life, AI Impact, and other structured categories. | Applies IAD, actors, incentives, rules, structural implications, and analytical reasoning. | Relates today's observation to prior state, arcs, reinforcement, discovery, and inflection. |
| **Variance Dimension** *What we assess when comparing execution engines* | **Evidence Variance** Did the engines find materially different evidence? | **Selection Variance** Given available evidence, did they disagree about significance? | **Classification Variance** Did they characterize the same development differently (Signal Type, Half-Life, AI role, etc.)? | **Interpretive Variance** Did they infer different inechanisms or meanings from substantially the same facts? | **Longitudinal Variance** Did they disagree about what the development means over time (e.g., discovery vs. reinforcement, arc assignment, inflection)? |
| **Examples of Questions** | • Different sources?<br>• Different geographic focus?<br>• Different events? | • Included vs. excluded?<br>• Borderline significance treated differently?<br>• Different story mix? | • Different Signal Type?<br>• Different Half-Life?<br>• Different AI Impact classification? | • Different IAD analysis?<br>• Different actor/incentive emphasis?<br>• Different structural implications? | • Discovery vs. reinforcement?<br>• Same arc or different arc?<br>• Inflection now or later?<br>• Different view of trajectory? |

## Key Insight

***"Disagreement upstream does not necessarily invalidate convergence downstream."***

Two execution engines may find different evidence, select different stories, or express different interpretations, yet still converge on the same higher-order analytical constructs — such as an institutional incentive shift, a long half-life signal, or a structural movement. The greater the depth of convergence (especially at the interpretation and longitudinal meaning stages), the stronger the evidence that the construct itself is robust and not an artifact of a particular execution engine.

## Dual Purpose of Provider B, C, D, etc.

Each additional provider serves two distinct purposes:

**1. Engineering Purpose**
Establish provider independence and remove any single provider as a point of dependency.

**2. Research Purpose**
Perturb the execution engine while holding the CyberBrief method substantially constant, then measure where analytical results vary and where higher-order constructs remain invariant.

## What This Enables

- Tests the durability of CyberBrief's analytical constructs.
- Identifies where model variance is natural, acceptable, or concerning.
- Improves the clarity and robustness of the CyberBrief method.
- Builds empirical confidence that the method is not model-dependent.
- Generates insights to strengthen analytical models (including IAD and others).
- Creates a basis for longitudinal research on analytical reliability.

## Scope of Application

The AVF can be applied to any CyberBrief analytical product, including:

- CyberBrief daily and retrospective briefs
- IAD analysis and outputs
- Models Ledger constructs
- Other Observatory research products
- Future analytical tools built on the CyberBrief method

*If an analytical construct remains stable across execution engines, it is stronger, more durable, and more trustworthy.*

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METHODS FOR A MORE STABLE WORLD

# CyberBrief Analytical Variance Framework (AVF)

*Different Engines. A Consistent Method. A Deeper Truth.*

SAME METHOD.
MULTIPLE PERSPECTIVES.
STRONGER INSIGHTS.

## Core Proposition

*"A durable analytical construct should remain meaningfully stable under changes in the execution engine, even when evidence discovery, selection, language, and local interpretation vary."*

The CyberBrief Analytical Variance Framework (AVF) provides a structured way to analyze where and why different execution engines (e.g., Claude, GPT, or others) agree or diverge when applying the same CyberBrief method to the same research problem.

### Why It Matters

- Greater confidence in findings
- More robust analytical models
- Method over model dependency
- Stronger, more resilient insights

## The Analytical Pipeline and Variance Points

| Analytical Pipeline<br>*What CyberBrief does at each stage* | 1. Evidence | 2. Selection | 3. Classification | 4. Interpretation | 5. Longitudinal Meaning |
|---|---|---|---|---|---|
| | Discovers and establishes the factual substrate. | Determines what is worthy of inclusion. | Assigns Signal Type, Half-Life, AI Impact, and other structured categories. | Applies IAD, actors, incentives, rules, structural implications, and analytical reasoning. | Relates today's observation to prior state, arcs, reinforcement, discovery, and inflection. |
| **Variance Dimension**<br>*What we assess when comparing execution engines* | **Evidence Variance**<br>Did the engines find materially different evidence? | **Selection Variance**<br>Given available evidence, did they disagree about significance? | **Classification Variance**<br>Did they characterize the same development differently (Signal Type, Half-Life, AI role, etc.)? | **Interpretive Variance**<br>Did they infer different mechanisms or meanings from substantially the same facts? | **Longitudinal Variance**<br>Did they disagree about what the development means over time? |
| **Examples of Questions** | • Different sources?<br>• Different geographic focus?<br>• Different events?<br>• Different depth of coverage? | • Included vs. excluded?<br>• Borderline significance treated differently?<br>• Different story mix?<br>• Different thresholds for inclusion? | • Different Signal Type?<br>• Different Half-Life?<br>• Different AI Impact classification?<br>• Different arc assignment? | • Different IAD analysis?<br>• Different actor/incentive emphasis?<br>• Different structural implications?<br>• Different causal framing? | • Discovery vs. reinforcement?<br>• Same arc or different arc?<br>• Inflection now or later?<br>• Different view of trajectory?<br>• Different long-term significance? |

**Variance can occur at any stage — upstream differences do not necessarily invalidate downstream convergence.**

## Potential Outcomes

### Convergence

Engines agree across the layer. Increases confidence in the construct.

### Partial Convergence

Some agreement, with meaningful differences. Signals areas for further analysis.

### Divergence

Engines disagree. Reveals sensitivity, underspecified criteria, or genuine interpretive difference.

### Construct Stability

Different evidence and reasoning paths, but convergence on higher-order constructs. Strongest evidence of robustness.

## Dual Purpose of Provider B, C, D, etc.

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### 2. Research Purpose

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## Applications

The AVF can be applied to any CyberBrief analytical product, including:

- CyberBrief daily and retrospective briefs
- IAD analysis and outputs
- Models Ledger constructs
- Other Observatory research products
- Future analytical tools built on the CyberBrief method

## Broader Relevance

The AVF helps evaluate whether key analytical products — not just CyberBrief — survive variance in the execution engine, including:

- IAD models (actors, incentives, arenas, rules)
- Longitudinal arc analysis
- Policy and governance assessments
- Threat and adversary trend models
- OSS ecosystem analysis
- Any structured analytical framework

## Guiding Principles

1. Same method. Different engines. Independent execution.
2. Variance is expected. It is a feature, not a failure.
3. Disagreement upstream does not invalidate convergence downstream.
4. The test is not whether another model can reproduce Claude, but whether the method survives a change of model.
5. Stable analytical constructs are more durable, more valuable, and more trustworthy.

# OSS Observatory – Conceptual Architecture

Daily cyber event monitoring, research interaction, durable record, and modeling — a human-in-the-loop system.

### Researcher (You)

- Runs daily CyberBrief production
- Discusses the brief and related topics
- Preserves selected conversations
- Moves promising thinking to modeling
- Explores the durable record
- Stays in control of what happens next

**Natural-language requests**

"Run today's brief."

"Preserve this discussion."

"Move this to modeling."
(and ordinary research questions)

**Responses, brief presentation, discussion, confirmation, results**

### LLM Research Partner (e.g., Claude)

- Interprets natural-language requests
- Presents the daily CyberBrief in the required format
- Engages in open-ended research discussion
- Helps identify and settle preservation boundaries
- Writes conversations to the durable record
- Supports exploration and modeling without creating premature artifacts

### 1. CyberBrief Production (Manual Run)

- Collects and processes daily cyber events and related material
- Produces the daily CyberBrief
- Writes output to the production location defined in `cyberbrief_production_workflow.md` (data/archive plane)
- Does not present to the screen

Produced CyberBrief (saved to disk)

### 2. Daily Discussion (LLM Interface)

- Loads and presents the completed CyberBrief for the current production day on request: `"Run today's brief."`
- Renders the brief in the screen-presentation format required by `cyberbrief_production_workflow.md` (Section 8.3)
- Supports open-ended research discussion about the brief and related topics using natural language

Selected discussion

Confirmation and provenance

### 3. Research Conversation Preservation

- Triggered by: "Preserve this discussion." (or equivalent)
- Identifies the coherent discussion and proposed boundary
- Confirms boundary (unless an explicit, unambiguous boundary is already provided)
- Writes the conversation substantially intact to `research_conversations/`
- Includes minimal provenance only

Relevant thinking

### 4. Modeling Bridge (Optional)

- Triggered by: "Move this to modeling." (or equivalent)
- Preserves relevant thinking for later continuation in a modeling context
- Does not declare or create a model, model candidate, Models Ledger entry, status, flag, classification, or other modeling artifact
- Keeps options open for future work

### 5. Persistent Storage (Repository)

Local file-based repository (this Git project)

#### CyberBrief Archive (data/archive plane)

- Daily CyberBrief output files
- Location and structure defined in `cyberbrief_production_workflow.md`
- Authoritative for where the current brief lives

#### Research Conversations
`research_conversations/`

- Preserved discussions, substantially intact
- Minimal provenance only
- No schema, index, scoring, tagging, or automated interpretation

#### Research Corpus
`research_corpus/`

- Source material and reference content used in research
- Growing collection
- Used to inform discussion and future work

#### Models (Future)
`models/`

- Future home for modeling work
- Not created by preservation or "Move this to modeling"
- Structure to be defined when earned by actual modeling work

#### Other Project Material

- Workflows and documentation (e.g., Architecture.md, `cyberbrief_production_workflow.md`)
- Prototypes, tests, and supporting tools (not part of core workflow today)

### 6. Governing Documentation (Authoritative References)

- Architecture.md — overall structure and interaction conventions (Section 8: Research Interaction Conventions)
- cyberbrief_production_workflow.md — authoritative for CyberBrief production, including where the current brief lives and how it is presented (Section 8.3)
- Other project documentation as needed (e.g., modeling, future components)